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NICO MACHAT, DANIEL KÖSTLER, MICHAEL SCHLEINZER

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1 Abstract

In this experiment, the method of nuclear magnetic resonance (NMR) was used to determine the relaxation times T_1 and T_2 . The parameter T_1 describes the longitudinal (spin–lattice) relaxation time, while T_2 characterizes the transverse (spin–spin) relaxation time. The aim of the experiment was the experimental determination of these two-time constants, as they are material-specific quantities and allow conclusions to be drawn about the composition of the investigated material.

2 Introduction

In this experiment, a sample of mineral oil is investigated. Neutrons and protons each carry a spin of $1/2$, and the composition of neutrons and protons within the atomic nucleus results in a total nuclear spin that is non-zero. Consequently, the atomic nucleus possesses a magnetic moment and aligns itself in an external magnetic field. The nuclear spin can be oriented either parallel or antiparallel to the magnetic field. Parallel alignment is energetically favorable and corresponds to a lower energy level, whereas antiparallel alignment corresponds to a higher energy level.

The investigated mineral oil interacts with its surrounding environment. Due to thermodynamic processes a slightly larger population of nuclei occupies the energetically favorable, parallel spin states compared to the antiparallel states. This population imbalance leads to a net magnetization of the material when placed in an external magnetic field.

The difference in the population of parallel and antiparallel spins is on the order of parts per million; however, this small imbalance is already sufficient to characterize the material.

Since spin can be described as a quantity that behaves analogously to angular momentum, a spin placed in a magnetic field begins to precess, similar to a spinning top in the Earth's magnetic field. This means that, in addition to aligning parallel or antiparallel to the external magnetic field, the nuclear spins also undergo a precessional motion around the field direction. This precession is characterized by the Larmor frequency, which denotes the frequency at which the spins precess in the external magnetic field.

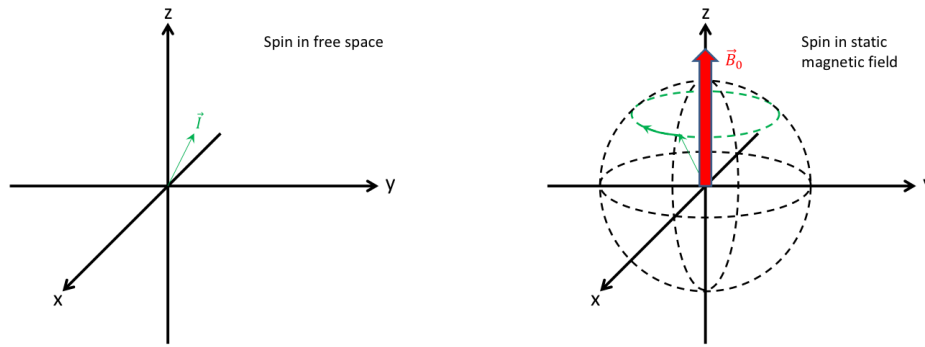


Figure 1 Schematic representation of a nuclear spin in free space and its precessional motion around the direction of an applied external magnetic field.

2.1 Longitudinal relaxation

The first effect is longitudinal relaxation, which is associated with an imbalance relative to the energetic equilibrium.

When the system is in thermal equilibrium, a larger fraction of spins is aligned parallel to the external magnetic field, corresponding to the energetically favorable state. If this equilibrium is disturbed, the system tends to return to the energetically optimal configuration. Such a disturbance can be achieved by applying a so-called π - pulse, which effectively alters the spin populations and drives the system out of equilibrium. As a result, a non-equilibrium population distribution is created, which does not correspond to the minimum energy state.

From this point onward, the system relaxes back toward thermal equilibrium. The process describing how the populations of parallel and antiparallel spins return to their equilibrium values is governed by an exponential time dependence characterized by the time constant T_1 . This time constant T_1 is referred to as the longitudinal (spin–lattice) relaxation time. The longitudinal relaxation time is a material-specific quantity, as it depends on the composition of the material and on the interactions between the nuclear spins and their surrounding environment.

In the following, the expression for the magnetization in the z-direction (which is also the direction of the external magnetic field) is given as a function of time, the time constant T_1 , and the equilibrium magnetization M_0 , which represents the magnetization in thermal equilibrium:

$$M_z = M_0(1 - e^{-t/T_1}).$$

2.2 Transverse relaxation

The second relevant effect is transverse relaxation. When a $\frac{\pi}{2}$ - pulse is applied to the sample, the nuclear spins are rotated from the z-axis into the xy-plane. In this process, spins initially aligned parallel to the positive z-direction are rotated into the positive y-direction, while spins initially aligned antiparallel to the magnetic field (negative z-direction) are rotated into the negative y-direction, when visualized on the Bloch sphere.

Due to the fact that the material is not perfectly homogeneous and that interactions between the nuclear spins are subject to stochastic fluctuations, the Larmor frequency is not identical for all nuclei.

Since the Larmor frequency depends on the local magnetic field, material-specific inhomogeneities lead to slightly different precession frequencies for different spins. As a consequence, some spins precess faster while others precess more slowly.

This causes the spins in the xy-plane to lose phase coherence over time, as they no longer precess synchronously around the z-axis. This process is referred to as dephasing. In addition to intrinsic material effects, magnetic field inhomogeneities and molecular motion further contribute to this loss of phase coherence, as spins experience different local magnetic fields over time.

The dephasing process that arises purely from interactions within the material is characterized by the transverse relaxation time T_2 . Experimentally, however, the observed decay also includes additional dephasing due to magnetic field inhomogeneities, which leads to an effective relaxation time denoted by T_2^* , with $T_2^* < T_2$. Both relaxation processes exhibit an exponential decay behavior. In the following, both T_2 and T_2^* will be determined experimentally, as their values provide further information about the material properties.

$$M_{trans} = M_0 \cdot e^{-t/T_2} \text{ \& \; } M_{trans} = M_0 \cdot e^{-t/T_2^*}$$

2.3 Spin Manipulation by the RF Field

When a radiofrequency (RF) field is applied to the sample, resonant coupling to the nuclear spins occurs only if the RF frequency matches the Larmor frequency. Under this resonance condition, the RF field efficiently interacts with the magnetic moments of the spins; for non-resonant frequencies, a detuning occurs, resulting in an oscillatory response of the magnetization. The RF field is generated by an alternating current in the RF coil (see Section 3, Experimental Setup) and produces a magnetic field component perpendicular to the static magnetic field B_0 . In a reference frame rotating at the Larmor frequency, this oscillating RF field appears as an effective static magnetic field B_1 , which governs the subsequent spin dynamics.

Due to the torque exerted by this effective field, the net magnetization begins to precess about the direction of B_1 . As a result, the magnetization is rotated away from the equilibrium orientation along the z-axis. The rotation angle depends on the strength of the RF field and the duration of the applied pulse. By adjusting the pulse length, defined rotations such as $\frac{\pi}{2}$ and π -pulses can be generated, allowing precise control of the spin orientation.

After the RF pulse is switched off, the effective field B_1 vanishes and the magnetization evolves freely in the laboratory frame. The transverse component of the magnetization then precesses around the static magnetic field B_0 at the Larmor frequency. This coherent precession of the transverse magnetization produces a time-dependent magnetic flux through the RF coil, which induces an oscillating voltage according to Faraday's law of induction. This induced voltage constitutes the measured NMR signal.

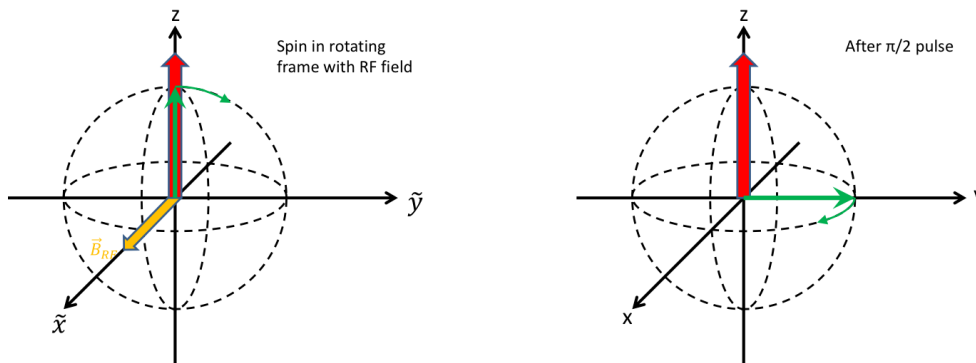


Figure 2 Schematic illustration of the spin dynamics in the rotating frame under the influence of an RF field and the resulting spin orientation after a $\frac{\pi}{2}$ -pulse.

3 Experimental setup

For this experiment, a permanent NdFeB magnet is provided, which generates the static magnetic field. In addition, a PS2 controller is used to adjust the magnetic field gradients in order to improve the homogeneity of the magnetic field. Linear gradient components along the x-, y-, and z-directions, as well as a quadratic component along the z-direction, can be controlled.

The experimental setup includes a receiver coil placed around the sample inside the static magnetic field of the permanent magnet. This coil serves two purposes. First, when driven by an alternating voltage, it generates an oscillating magnetic field, which forms the basis for the RF pulses applied to the sample. Second, the same coil is used to detect the magnetization of the sample, as the time-dependent transverse magnetization induces a voltage in the coil.

The induced signal is processed by the receiver unit, which amplifies and conditions the signal and forwards it to the oscilloscope, where it can be recorded and analyzed.

The RF signal itself is generated by a synthesizer, which produces a high-frequency alternating voltage in the megahertz range, close to the Larmor frequency of the nuclear spins. A pulse programmer controls the temporal structure of this signal by defining when and for how long the RF voltage is applied to the coil. In this way, RF pulses with well-defined durations can be generated, allowing precise control of the applied pulse sequences. The analysis of the applied pulses and the observation of the resulting signals are performed using an oscilloscope.

Figure 3 shows the unit comprising the receiver, synthesizer, and pulse programmer.

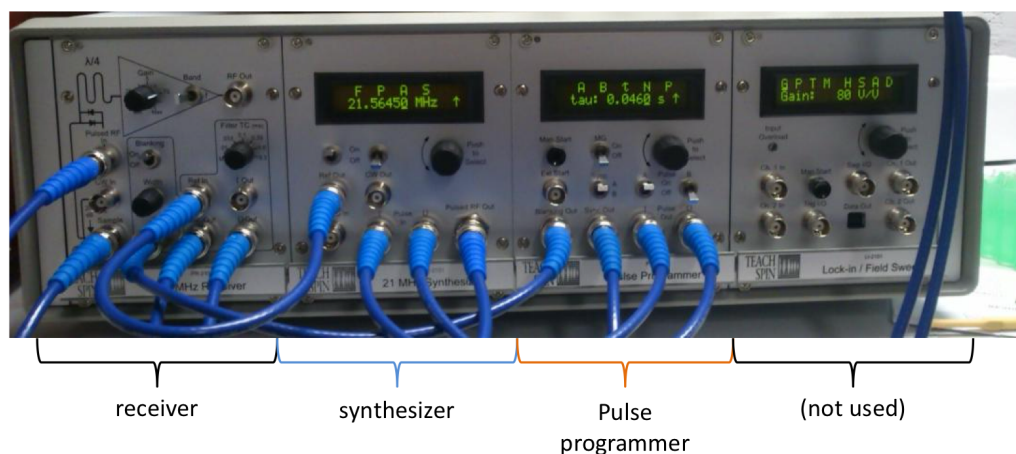


Figure 3 RF-pulses- unit

4 Experimental Procedure

In the following, the procedures used to determine the longitudinal relaxation time T_1 and the transverse relaxation times T_2 and T_2^* are described.

4.1 Determination of the Resonance Frequency

As a first step, the resonance frequency of the experimental setup was determined. As described in Section 2.3, the frequency of the applied RF field must be resonant with the Larmor frequency of the nuclear spins. To identify this resonance frequency, the duration of pulse A was initially set to $1\mu\text{s}$ and the repetition period was set to 300 ms.

The signals from Channel 1 and Channel 2 were then displayed on the oscilloscope. Channel 1 represents the signal envelope and therefore only shows the amplitude of the detected signal. As it displays the absolute value of the signal, it is always positive and indicates both the response to the RF pulse and the voltage induced by the precessing magnetization.

Channel 2 provides phase-sensitive information and indicates the direction of the induced voltage.

After optimizing the display settings, an oscillation was observed in the signal of Channel 2. As explained in Section 2.3, this oscillation arises from a frequency mismatch between the applied RF field and the Larmor frequency. To reach resonance, the frequency of the synthesizer was varied until this oscillation disappeared. The disappearance of the oscillations indicates that the RF frequency matches the Larmor frequency, at which point a stable and clearly observable signal was obtained.

The resonance frequency determined in this procedure was:

$$f_{res} = 21.53025 \text{ MHz.}$$

Using the measured resonance frequency, the magnitude of the static magnetic field B_0 was calculated using the gyromagnetic ratio of the proton. Since the NMR signal in the investigated mineral oil originates predominantly from hydrogen nuclei, the proton gyromagnetic ratio was used. The magnetic field strength was determined via the Larmor relation

$$\omega_0 = \gamma \cdot B_0 \text{ \& } \gamma = 2.675 \cdot 10^8 \frac{\text{rad}}{\text{sT}} \rightarrow B_0 = \frac{2\pi f_{res}}{\gamma} = 0.5057 \text{ T}$$

4.2 Fine Tuning of the Magnetic Field

In the next step, the magnetic field was fine-tuned, as it is not sufficiently homogeneous by default. This adjustment was performed using the PS2 controller, as described in Section 3. The magnetic field gradients were varied using the control knobs until the signal on Channel 1 remained significantly greater than zero for the longest possible duration.

During this optimization process, it was observed that the signal on Channel 2 began to exhibit oscillations again. This behavior indicates that the resonance condition achieved in the previous step was disturbed by the changes in the magnetic field. Therefore, the optimization was carried out iteratively: first, the magnetic field was adjusted to maximize the duration of the Channel 1 signal, and subsequently, the synthesizer frequency was fine-tuned once more until the oscillations in Channel 2 disappeared again. In the figure below, the signal of Channel 1 is shown in yellow, while the signal of Channel 2 is shown in green.

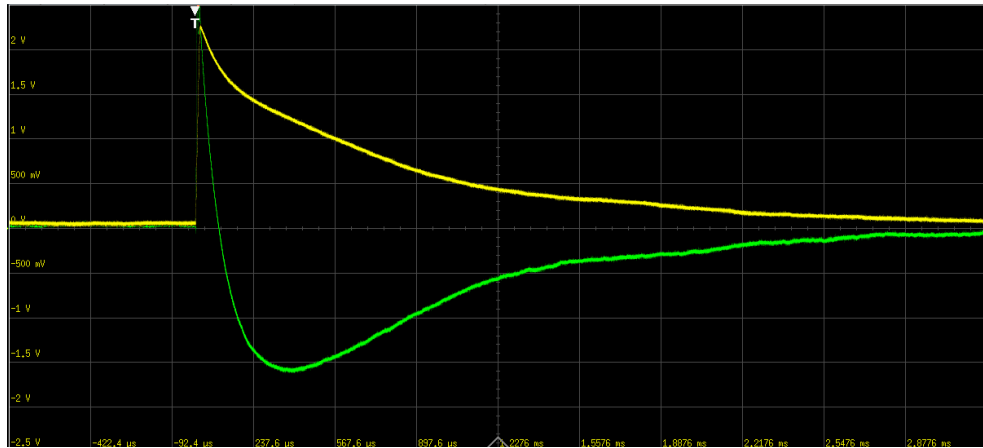


Figure 4 Signal of Channel 1 and Channel 2 after fine tuning

As a consequence of the magnetic field fine-tuning procedure, the previously adjusted resonance condition was slightly altered. The modification of the magnetic field led to a corresponding change in the resonance frequency, as the RF frequency was not automatically readjusted during this process. Therefore, after fine tuning of the magnetic field, new values for both the resonance frequency and the static magnetic field strength B_0 were obtained.

$$f_{res} = 21.52798 \text{ MHz.}$$

$$B_0 = \frac{2\pi f_{res}}{\gamma} = 0.5055 \text{ T}$$

$$\Delta B = 0.0002 \text{ T}$$

4.3 Calibration of the RF Pulse Lengths

As described in Section 2.3, the duration of the RF pulses must be calibrated in order to generate defined pulse angles such as $\frac{\pi}{2}$, π , $\frac{3\pi}{2}$, and 2π - pulses. For this purpose, the duration of pulse A was varied from 0 μs to 20 μs in steps of 1 μs using the pulse programmer.

For each pulse duration, the signal amplitudes measured on Channel 1 and Channel 2 were recorded (Table 1). The signal on Channel 2 is phase-sensitive and therefore exhibits positive and negative values, resulting in a sinusoidal dependence on the pulse duration. In contrast, the signal on Channel 1 represents the signal envelope and is always positive but still follows a sinusoidal modulation.

A sinus fit was applied to the recorded data. From this fit, the characteristic pulse durations were identified. The first maximum corresponds to a $\frac{\pi}{2}$ - pulse, the first zero crossing corresponds to a π - pulse, the second maximum corresponds to a $\frac{3\pi}{2}$ - pulse, and the second zero crossing corresponds to a 2π - pulse.

This behavior can be explained by the orientation of the spins after each pulse. A $\frac{\pi}{2}$ - pulse rotates the magnetization entirely into the transverse plane, resulting in the maximum induced voltage in the detection coil and therefore the largest signal amplitude. A π - pulse inverts the spin populations along the z-axis, leaving no transverse magnetization and thus yielding a minimum signal. A $\frac{3\pi}{2}$ - pulse again produces transverse magnetization, but with opposite orientation, leading to a signal with opposite sign. This sequence explains the observed sinus dependence of the signal amplitude on the pulse duration.

With the software “Igor Pro” the datapoints were plotted in a diagram and a sinusoidal fit was applied to it (Figure 5 and 6). The pulse durations corresponding to the individual pulse angles were read out of the fit with the software:

$\frac{\pi}{2}$ - pulse: 4.9397 μs

π - pulse: 10.261 μs

$\frac{3\pi}{2}$ - pulse: 16.084 μs

The 2π - pulse could not be determined, since the maximum pulse duration of 20 μs was insufficient to reach the second zero crossing.

A [μ s]	CH1 [V]	CH2 [V]
0.02	0	0
1	1.35	-0.100
2	2.27	-0.194
3	2.93	-0.262
4	3.42	-0.305
5	3.67	-0.320
6	3.69	-0.310
7	3.46	-0.271
8	3.04	-0.212
9	2.48	-0.134
10	1.83	-0.100
11	1.28	0.130
12	1.40	0.211
13	1.83	0.287
14	2.24	0.347
15	2.56	0.383
16	2.74	0.398
17	2.72	0.384
18	2.51	0.349
19	2.14	0.289
20	1.63	0.216

Table 1 Pulse duration and corresponding signal amplitudes for Channel 1 and Channel 2

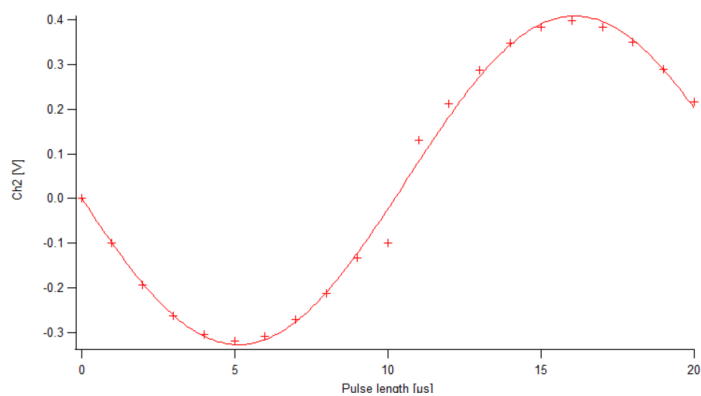


Figure 5 Channel 2 signal as a function of pulse duration with sinus fit.

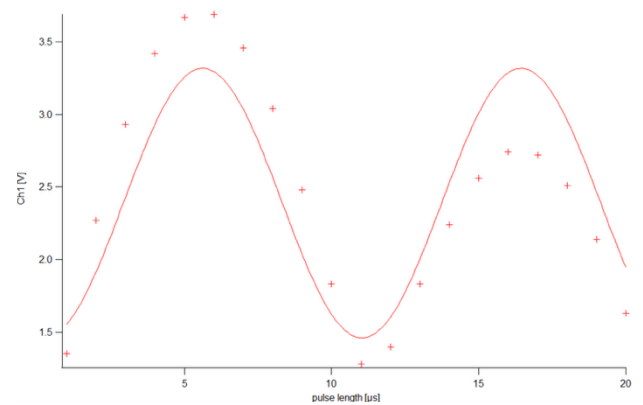


Figure 6 Channel 1 signal as a function of pulse duration with sinus fit.

4.4 Determination of the Transverse Relaxation Time T_2^*

To determine the inhomogeneous transverse relaxation time T_2^* , a $\frac{\pi}{2}$ -pulse was applied to the sample, as described in Section 2.2. The pulse length A on the pulse programmer was set to the previously determined duration corresponding to a $\frac{\pi}{2}$ -pulse.

After application of the pulse, the decay of the transverse magnetization was recorded using Channel 1 of the oscilloscope. As discussed in Section 2.2, the observed signal exhibits an exponential decay due to dephasing effects. By fitting an exponential function to this decay, the time constant T_2^* was determined.

The measured signal was exported from the oscilloscope as a CSV file and subsequently analyzed using a Python script. An exponential fit was applied to the data in the relevant time interval. Initial data points were excluded from the fit, as they do not correspond to the relaxation decay. From the fitted exponential function, the inhomogeneous transverse relaxation time T_2^* was determined. The corresponding plot is shown in Figure 7.

The value obtained for the inhomogeneous transverse relaxation time was:

$$T_2^* = 1.09 \text{ ms.}$$

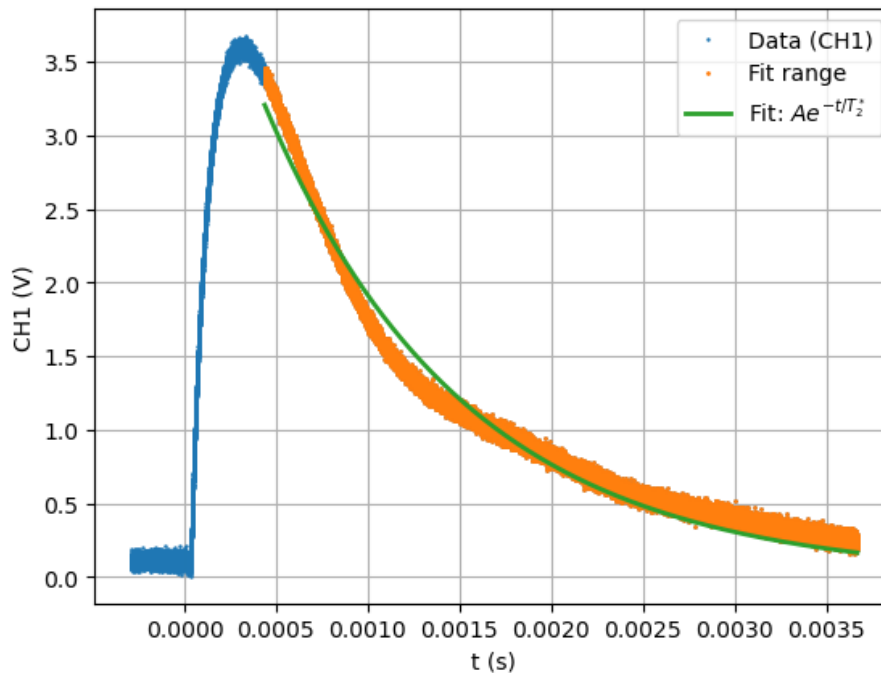


Figure 7 Channel 1 signal envelope with exponential fit for the determination of T_2^*

4.5 Determination of the Longitudinal Relaxation Time T_1

To determine the longitudinal relaxation time T_1 , a π - pulse was applied to the sample, as described in Section 2.1. This pulse inverts the spin populations, driving the system out of thermal equilibrium by creating an excess of spins aligned antiparallel to the static magnetic field. The system subsequently relaxes back toward equilibrium, where a slight excess of parallel spins is present.

The relaxation process was probed by applying a $\frac{\pi}{2}$ - pulse after a variable delay time τ . This pulse rotates the longitudinal magnetization into the transverse plane, allowing the magnetization at time τ to be detected via the induced signal in the RF coil. Immediately after the π - pulse, a large signal with opposite sign compared to equilibrium is observed. As the delay time τ increases, the signal amplitude decreases, crosses zero when the populations of parallel and antiparallel spins are equal, and subsequently approaches the equilibrium value.

By repeating this measurement for different values of τ and fitting the resulting data with an exponential function, the longitudinal relaxation time T_1 was extracted.

By solving the given differential equation, the longitudinal relaxation time T_1 can be determined:

$$\frac{dM_z(t)}{dt} = -\frac{1}{T_1}(M_z(t) - M_0)$$

$$\frac{dM_z}{M_z - M_0} = -\frac{dt}{T_1}$$

$$\int \frac{dM_z}{M_z - M_0} = -\frac{1}{T_1} \int dt$$

$$\ln|M_z - M_0| = -\frac{t}{T_1} + C$$

$$M_z - M_0 = C e^{-t/T_1}$$

$$M_z(t) = M_0 + C e^{-t/T_1}$$

In this case, the longitudinal magnetization at time $t = 0$ is given by $M_z(0) = -M_0$, since the spin populations are completely inverted by the π - pulse. As a result, the magnetization has the same magnitude as in equilibrium, but with opposite sign. Inserting this initial condition into the equation yields:

$$-M_0 = M_0 + C$$

$$C = -2M_0$$

$$M_z(t) = M_0 - 2M_0 e^{-t/T_1}$$

The time at which the magnetization crosses zero, corresponding to an equal population of parallel and antiparallel spins and therefore no detectable signal, is denoted by t_0 . At this time, the longitudinal magnetization must vanish, since the contributions of the spins cancel each other. This leads to the following equation:

$$M_z(t_0) = 0$$

$$0 = 1 - 2e^{-t_0/T_1}$$

$$e^{-t_0/T_1} = \frac{1}{2}$$

$$T_1 = \frac{t_0}{\ln 2}$$

The zero-crossing time was determined to be $t_0 = 25$ ms, as shown in Table 2. Using the relation above the longitudinal relaxation time was calculated to be

$$T_1 = 36 \text{ ms}$$

In addition, the measured data listed in Table 2 were plotted and fitted using Python according to the derived model function (Figure 8). From this fit, a value of $T_1 = 39.4 \text{ ms}$ was obtained, showing a slight deviation from the T_1 value determined previously.

τ [s]	Ch2 [V]
0,001	0,403
0,002	0,343
0,003	0,310
0,004	0,280
0,005	0,270
0,006	0,253
0,007	0,234
0,008	0,217
0,009	0,200
0,01	0,181
0,011	0,165
0,012	0,145
0,013	0,135
0,014	0,121
0,015	0,108
0,016	0,093
0,017	0,081
0,018	0,070
0,019	0,060
0,020	0,049
0,021	0,026
0,022	0,021
0,023	0,014
0,024	0,008
0,025	0
0,030	-0,025
0,035	-0,037
0,040	-0,046
0,045	-0,057
0,050	-0,064

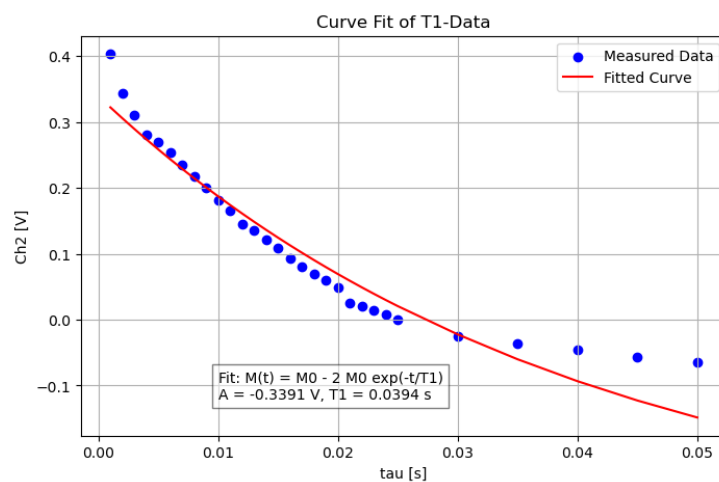
Table 2 Measured signal amplitudes of Channel 2 as a function of the delay time τ 

Figure 8 Curve fit of the Data for T1

4.6 Determination of the Transverse Relaxation Time T_2

In the following step, the true transverse relaxation time T_2 was determined. As discussed in Section 2.2, the previously measured T_2^* does not represent the intrinsic transverse relaxation time, since it also includes dephasing effects caused by inhomogeneities of the static magnetic field, which lead to different precession frequencies of the spins.

To compensate for these magnetic field inhomogeneities, a Hahn echo sequence was applied. First, a $\frac{\pi}{2}$ -pulse was used to rotate the magnetization into the transverse plane. After the signal had decayed to approximately 10% of its maximum value, a π -pulse was applied. Prior to the π -pulse, the spins dephase due to different precession frequencies. The π -pulse effectively inverts the phase distribution, causing the spins to rephase and form an echo signal after an additional delay time τ .

While the π -pulse compensates for dephasing caused by static magnetic field inhomogeneities, it does not reverse dephasing originating from material-specific stochastic fluctuations. These fluctuations are random in nature and therefore continue to contribute to an irreversible loss of phase coherence. Additionally, molecular motion causes spins to sample different regions of the inhomogeneous magnetic field over time, which cannot be fully compensated by the echo sequence.

By varying the delay time τ between the $\frac{\pi}{2}$ - and π -pulses, the amplitude of the Hahn echo was measured for different values of τ . For shorter delay times, less irreversible dephasing occurs, resulting in a larger echo amplitude, whereas for longer delay times increased dephasing leads to a reduced echo signal. The decay of the echo amplitude as a function of τ follows an exponential behavior, from which the true transverse relaxation time T_2 was determined.

The data were analyzed using Python, and an exponential fit was applied to the measured data points, yielding a transverse relaxation time of

$$T_2 = 12.362 \text{ ms.}$$

The fit is shown in Figure 9, and the corresponding data are listed in Table 3.

τ [s]	Ch1 [V]
0,002	1,10
0,004	0,90
0,006	0,75
0,008	0,64
0,010	0,55
0,012	0,47
0,014	0,41
0,016	0,36
0,018	0,31
0,020	0,27

Table 3 Measured Hahn echo amplitudes for different delay times τ .

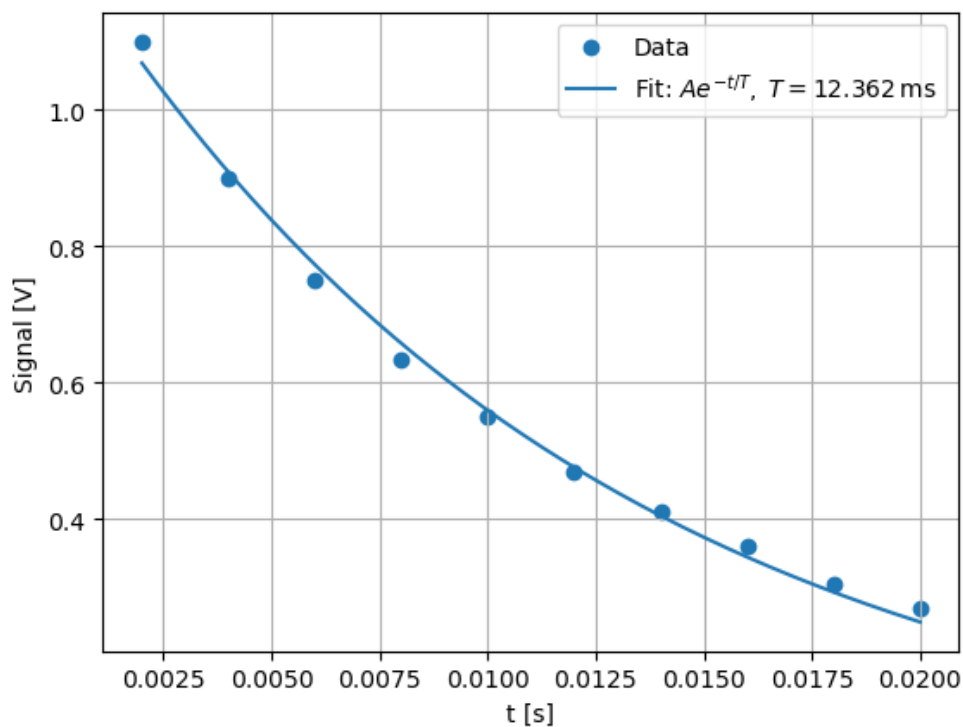


Figure 9 Hahn echo amplitude as a function of delay time τ with exponential fit used to determine T_2

The magnetic field inhomogeneity ΔB_0 was determined using the following relation:

$$\frac{1}{T_2} = \frac{1}{T_2^*} - \frac{\gamma \Delta B_0}{2} \rightarrow \Delta B_0 = \frac{2}{\gamma} \left(\frac{1}{T_2^*} - \frac{1}{T_2} \right) = \frac{2}{\gamma} \left(\frac{1}{0.75 \text{ ms}} - \frac{1}{12.362 \text{ ms}} \right) \rightarrow \Delta B_0 = 9.364 \mu\text{T}$$

4.7 Determination of T_2 Using Multiple-Pulse and CPMG Sequences

4.7.1 Multiple Pulse (CP-sequence)

The difficulty in determining T_2 in the previous approach is that while the method works reasonably well for small delay times τ , the time interval during which the spins can dephase increases significantly for larger τ . As this interval becomes longer, it becomes increasingly unlikely, due to molecular motion as well as variations in the magnetic field inhomogeneity, that, after application of the π - pulse, the spins retrace their paths through exactly the same field inhomogeneities in an opposite phase. Consequently, the refocusing becomes incomplete. It would therefore be advantageous to reduce this time interval.

To achieve this, a $\frac{\pi}{2}$ - pulse is applied, followed by a delay time τ , then a π - pulse, another delay time τ , and subsequently additional π - pulses applied at equal time intervals 2τ . In contrast to the previous method, where only a single π - pulse was used, multiple π - pulses are now applied sequentially. This effectively minimizes the dephasing intervals, such that even for large overall evolution times the individual dephasing periods remain short. The shorter these time intervals are, the more likely it is that the spins experience the same magnetic field inhomogeneities before and after phase inversion. As a result, this method provides a significantly more accurate determination of T_2 compared to measurements based on single π - pulse sequences. Figure 10 clearly shows the exponential decay associated with the transverse relaxation time T_2 .

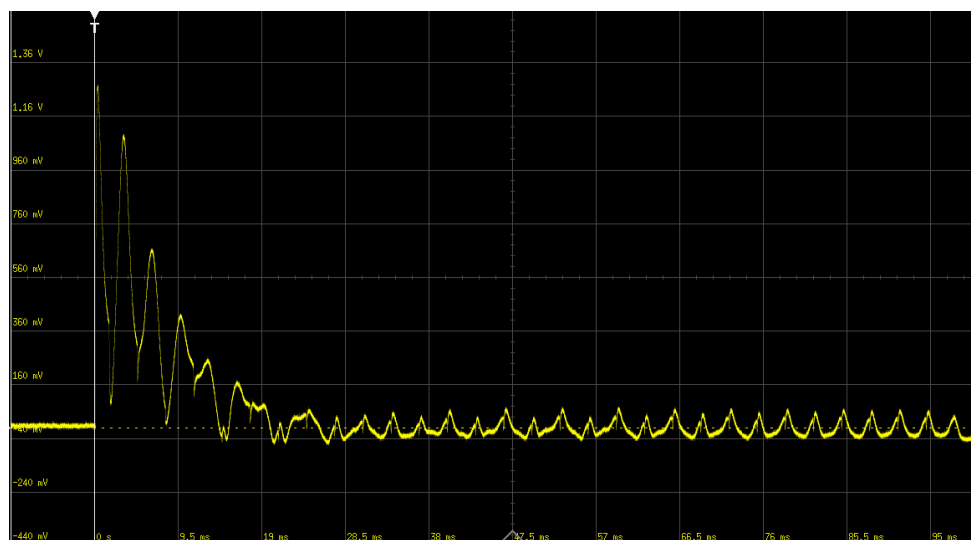


Figure 10 Echo train obtained using the CP pulse sequence: $\frac{\pi}{2} - \tau - \pi - 2\tau - \pi - \dots$

4.7.2 CPMG-Sequence (Carr-Purcell-Meiboom-Gill)

In the previous method, an additional limitation arises from the fact that the applied RF pulses, particularly the π - pulses, but also the $\frac{\pi}{2}$ - pulses, are not perfectly accurate in practice. Instead of producing an exact rotation of 180° , the pulses typically exhibit small flip-angle errors, for example rotating the magnetization by 178° . While such an error is negligible for a single pulse, it becomes problematic when the pulse sequence is repeated multiple times. In this case, the errors accumulate and lead to the build-up of an unwanted magnetization component along the z-direction.

This problem can be mitigated by modifying the pulse sequence according to the Meiboom–Gill approach. The $\frac{\pi}{2}$ - pulse is applied about one transverse axis (e.g. the x-axis), while the subsequent π - pulses are applied about the orthogonal transverse axis (e.g. the y-axis). As a result, flip-angle errors introduced by the π - pulses do not accumulate coherently.

In the Bloch-sphere picture, the initial $\frac{\pi}{2}$ - pulse is applied about the x-axis, rotating the magnetization into a transverse plane, ideally the xy-plane but potentially with a small residual z-component due to pulse imperfections. The spins then evolve freely during a time interval τ , after which a π - pulse is applied about the orthogonal y-axis. Following another interval τ , a spin echo is formed, and the magnetization continues to evolve for an additional time τ . At this point, when the next π - pulse is applied, each magnetization vector is located almost at the same position on the Bloch sphere as during the previous π - pulse.

Consequently, any flip-angle error introduced by the preceding π - pulse is compensated to first order by the subsequent pulse. As a result, pulse errors do not accumulate coherently: one echo may exhibit a small out-of-plane component corresponding to the error of a single π - pulse, while the subsequent echo lies predominantly in the plane defined by the initial $\frac{\pi}{2}$ - pulse.

From Figure 11 it is clearly evident that the signal evolution is significantly smoother and more stable than in the CP- sequence.

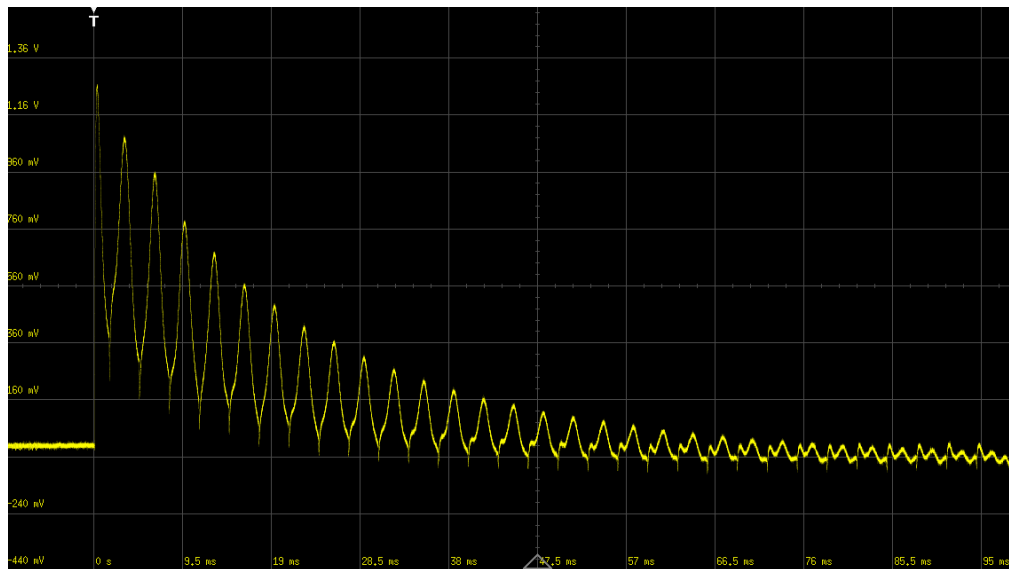


Figure 11 Echo train obtained using the CPMG pulse sequence

4.7.3 Data Analysis and Comparison of CP and CPMG

Using the CSV data acquired from the CPMG pulse sequence, an exponential fit was performed to determine T_2 . The fit is shown in Figure 12. The resulting transverse relaxation time is

$$T_2 = 19.459 \text{ ms.}$$

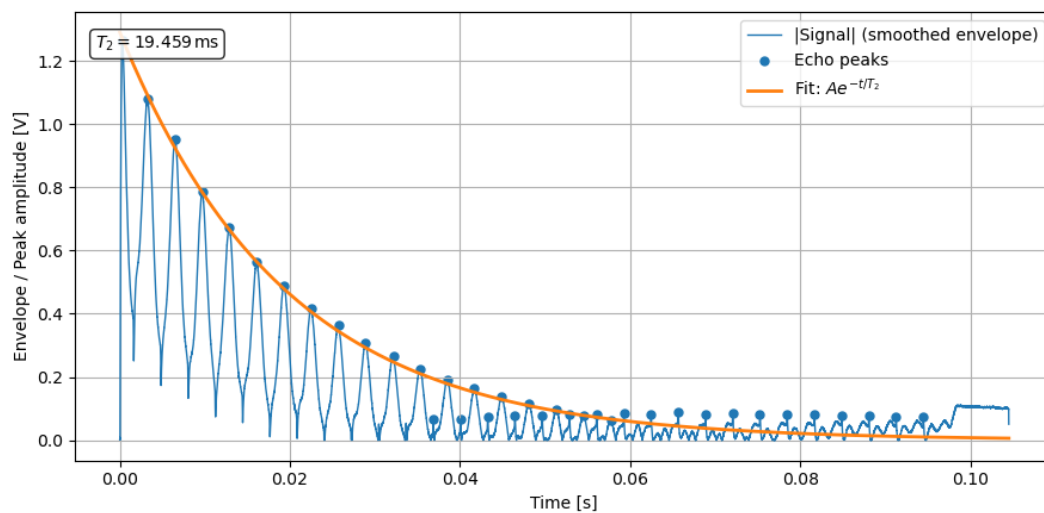


Figure 12 Exponential fit of the CPMG echo envelope used to determine the transverse relaxation

Using the CSV data acquired from the CP pulse sequence, an exponential fit was performed to determine T_2 . The fit is shown in Figure 13. The resulting transverse relaxation time is

$$T_2 = 7.31 \text{ ms.}$$

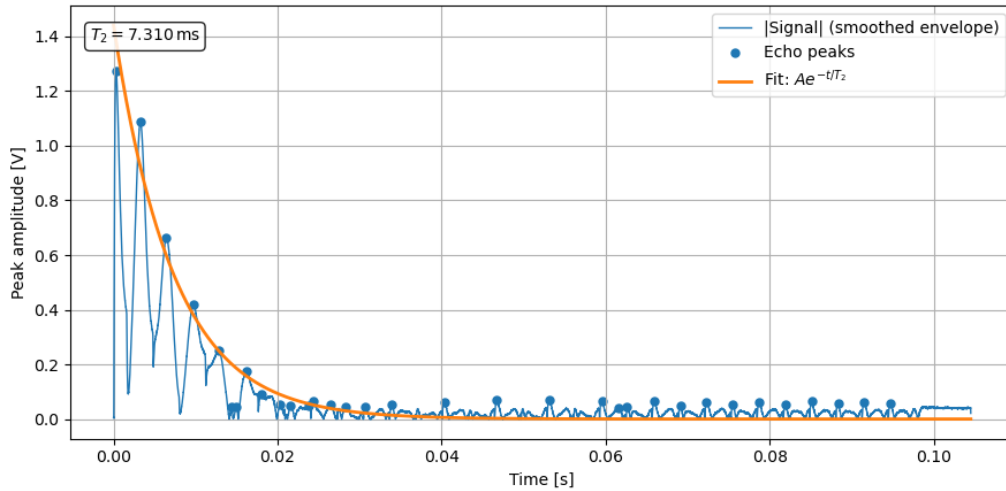


Figure 13 Exponential fit of the CP echo envelope used to determine the transverse relaxation

The significantly shorter T_2 obtained from the CP sequence compared to both the single-echo and CPMG measurements (see next chapter) indicates substantial imperfections of the π -pulses. In the CP sequence, these pulse errors accumulate coherently, leading to an artificial and rapid decay of the echo train, which is clearly visible in the measured signal. The CPMG sequence suppresses this error accumulation, yielding a larger and more reliable estimate of the intrinsic T_2 .

The enhanced stability of the CPMG sequence compared to the CP sequence can be demonstrated by deliberately detuning the π -pulse duration.

In our experiment, the π -pulse was increased from 10.261 μs to 15 μs , representing a large flip-angle error close to a $\frac{3\pi}{2}$ -pulse. Despite this strong deviation, the CPMG signal remains relatively stable, as shown in Figure 15.

In contrast, the CP sequence rapidly collapses under the same conditions (Figure 14), since flip-angle errors accumulate coherently from pulse to pulse. In the CPMG sequence, however, these errors remain bounded and do not build up, resulting in a significantly more stable echo train. Only for pulse lengths approaching 16 μs does the CPMG sequence also begin to fail.

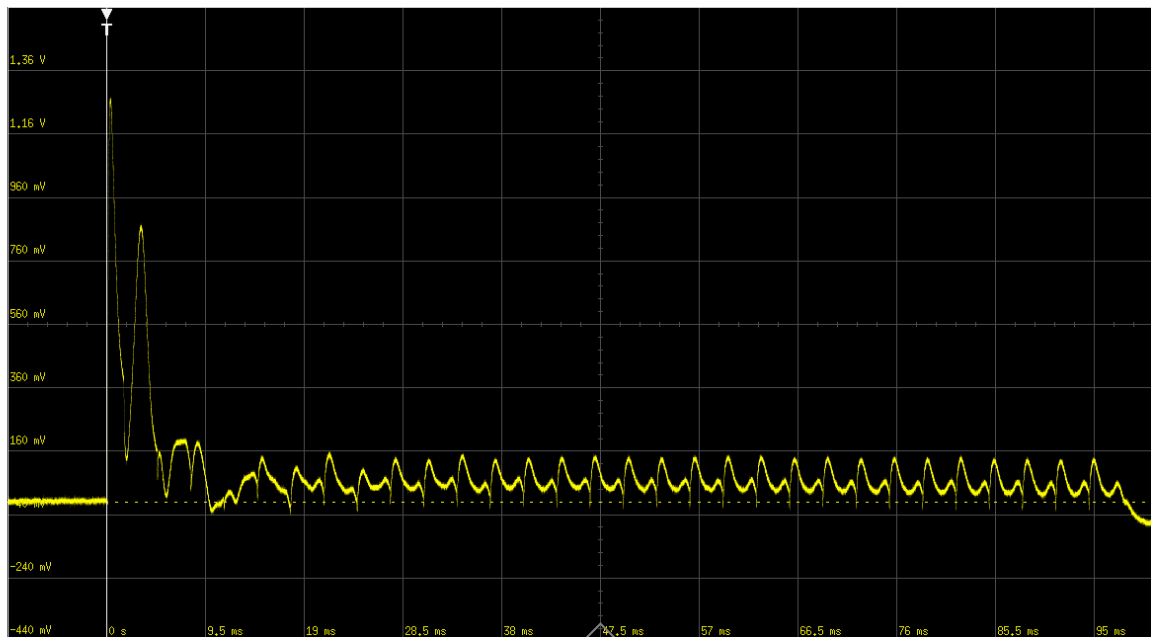


Figure 14 Recorded signal using the CP pulse sequence with a strongly detuned π - pulse duration

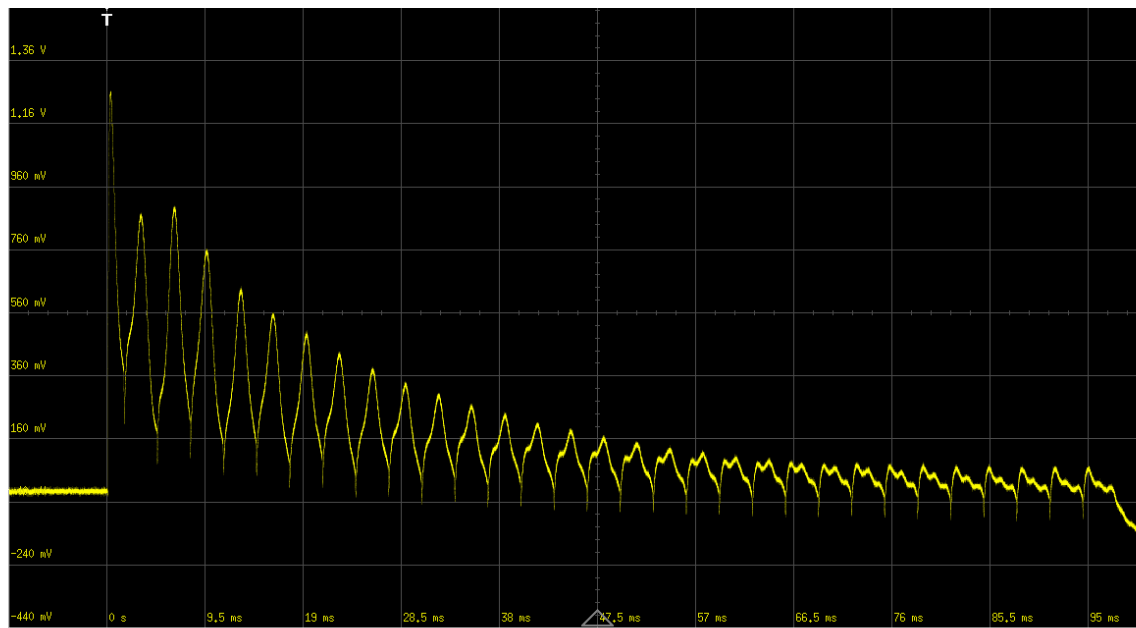


Figure 15 Recorded signal using the CPMG pulse sequence with a strongly detuned π - pulse duration

5 Conclusion and Interpretation

In summary, the experimental results are in good agreement with the theoretical expectations. The transverse relaxation time T_2^* was determined to be 1.09 ms, which is expected to be smaller than T_2 due to additional dephasing caused by magnetic field inhomogeneities. Accordingly, $T_2^* < T_2$ was clearly confirmed.

Using the single-pulse method, T_2 was first determined to be 12.362 ms, more than an order of magnitude larger than T_2^* . This large difference further indicates that the static magnetic field in the experimental setup was strongly inhomogeneous. Applying the CPMG method, T_2 was measured to be 19.459 ms, approximately twice the value obtained with the single-pulse method. This increase can be explained by the improved compensation of magnetic field inhomogeneities and pulse imperfections provided by the CPMG sequence. The measurement using the Carr–Purcell (CP) sequence without Meiboom–Gill phase cycling does not yield reliable results, as the extracted time constant T_2 is significantly smaller than the value obtained from the single-echo measurement. This indicates that the applied π - pulses suffer from substantial imperfections, whose errors accumulate coherently in the CP sequence and lead to an artificially accelerated decay of the echo train.

The longitudinal relaxation time T_1 was determined to be 36 ms, which is more than twice as large as the T_2 value obtained from the CPMG measurement. This relationship is consistent with theoretical expectations, further confirming the validity of the experimental results.

6 Appendix

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